Compute in the cloud

Amazon Elastic Compute Cloud (Amazon EC2)

- EC2 runs on top of physical host machines managed by AWS using virtualization technology.
- This idea of sharing underlying hardware is called multitenancy.
- The hypervisor is responsible for coordinating this multitenancy and it is managed by AWS.
- The hypervisor is responsible for isolating the virtual machines from each other as they share resources from the host.
- one EC2 instance is not aware of any other EC2 instances also on that host.

How Amazon EC2 works?

1. **Launch:** First, you launch an instance. Begin by selecting a template with basic configurations for your instance. These configurations include the operating system, application server, or applications. You also select the instance type, which is the specific hardware configuration of your instance.

   As you are preparing to launch an instance, you specify security settings to control the network traffic that can flow into and out of your instance. Later in this course, we will explore Amazon EC2 security features in greater detail.

2. **Connect**: connect to the instance. You can connect to the instance in several ways. Your programs and applications have multiple different methods to connect directly to the instance and exchange data. Users can also connect to the instance by logging in and accessing the computer desktop.

3. **Use**: After you have connected to the instance, you can begin using it. You can run commands to install software, add storage, copy and organize files, and more.

EC2 instance types

▼ General purpose

provide a balance of compute, memory, and networking resources. You can use them for a variety of workloads, such as:

- application server
- gaming server
- backend server for enterprise application
- small and medium database

▼ Compute optimized

ideal for compute-bound applications that benefit from high-performance processors. Like general purpose instances, you can use compute optimized instances for workloads such as web, application, and gaming servers.

the difference is compute optimized applications are ideal for high-performance web servers, compute-intensive applications servers, and dedicated gaming servers. You can also use compute optimized instances for batch processing workloads that require processing many transactions in a single group.

▼ Memory optimized

designed to deliver fast performance for workloads that process large datasets in memory.

▼ Accelerated computing

use hardware accelerators, or coprocessors, to perform some functions more efficiently than is possible in software running on CPUs. Examples of these functions include floating-point number calculations, graphics processing, and data pattern matching.

Accelerated computing instances are ideal for workloads such as graphics applications, game streaming, and application streaming.

▼ Storage optimized

designed for workloads that require high, sequential read and write access to large datasets on local storage.

Examples of workloads suitable for storage optimized instances include distributed file systems, data warehousing applications, and high-frequency online transaction processing (OLTP) systems.

EC2 pricing

▼ On-Demand

ideal for short-term, irregular workloads that cannot be interrupted. No upfront costs or minimum contracts apply. The instances run continuously until you stop them, and you pay for only the compute time you use.

Sample use cases for On-Demand Instances include developing and testing applications and running applications that have unpredictable usage patterns.

On-Demand Instances are not recommended for workloads that last a year or longer because these workloads can experience greater cost savings using Reserved Instances.

▼ EC2 saving plans

reduce your compute costs by committing to a consistent amount of compute usage for a 1-year or 3-year term. This term commitment results in savings of up to 66% over On-Demand costs.

Any usage up to the commitment is charged at the discounted plan rate (for example, $10 an hour). Any usage beyond the commitment is charged at regular On-Demand rates.

▼ Reserved Instances

billing discount applied to the use of On-Demand Instances in your account.

You can purchase Standard Reserved and Convertible Reserved Instances for a 1-year or 3-year term, and Scheduled Reserved Instances for a 1-year term. You realize greater cost savings with the 3-year option.

At the end of a Reserved Instance term, you can continue using the Amazon EC2 instance without interruption. However, you are charged On-Demand rates until you do one of the following:

- Terminate the instance.
- Purchase a new Reserved Instance that matches the instance attributes (instance type, Region, tenancy, and platform).

▼ Spot instances

ideal for workloads with flexible start and end times, or that can withstand interruptions. Spot Instances use unused Amazon EC2 computing capacity and offer you cost savings at up to 90% off of On-Demand prices.

example:

> Suppose that you have a background processing job that can start and stop as needed (such as the data processing job for a customer survey). You want to start and stop the processing job without affecting the overall operations of your business. If you make a Spot request and Amazon EC2 capacity is available, your Spot Instance launches. However, if you make a Spot request and Amazon EC2 capacity is unavailable, the request is not successful until capacity becomes available. The unavailable capacity might delay the launch of your background processing job. After you have launched a Spot Instance, if capacity is no longer available or demand for Spot Instances increases, your instance may be interrupted. This might not pose any issues for your background processing job. However, in the earlier example of developing and testing applications, you would most likely want to avoid unexpected interruptions. Therefore, choose a different EC2 instance type that is ideal for those tasks.

▼ Dedicated hosts

physical servers with Amazon EC2 instance capacity that is fully dedicated to your use.

You can use your existing per-socket, per-core, or per-VM software licenses to help maintain license compliance. You can purchase On-Demand Dedicated Hosts and Dedicated Hosts Reservations. Of all the Amazon EC2 options that were covered, Dedicated Hosts are the most expensive.

Scaling EC2

Scalability involves beginning with only the resources you need and designing your architecture to automatically respond to changing demand by scaling out or in.

If you wanted the scaling process to happen automatically, which AWS service would you use? The AWS service that provides this functionality for Amazon EC2 instances is **Amazon EC2 Auto Scaling**.

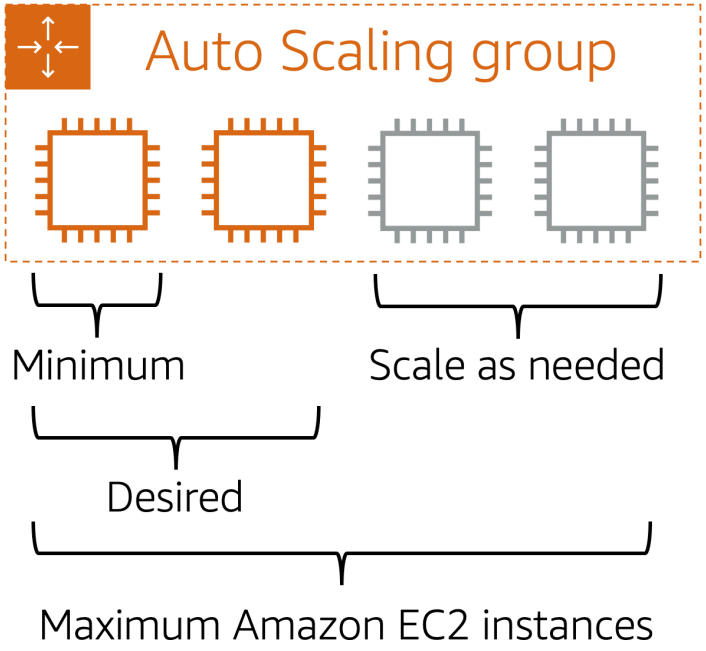
Amazon EC2 Auto scaling

Amazon EC2 Auto Scaling enables you to automatically add or remove Amazon EC2 instances in response to changing application demand.

By automatically scaling your instances in and out as needed, you are able to maintain a greater sense of application availability.

Within Amazon EC2 Auto Scaling, you can use two approaches: dynamic scaling and predictive scaling.

- *Dynamic scaling* responds to changing demand.
- *Predictive scaling* automatically schedules the right number of Amazon EC2 instances based on predicted demand.



Next, you can set the **desired capacity** at two Amazon EC2 instances even though your application needs a minimum of a single Amazon EC2 instance to run.

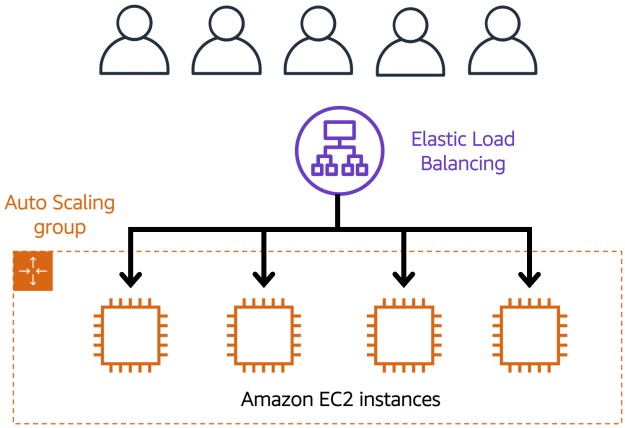
If you do not specify the desired number of Amazon EC2 instances in an Auto Scaling group, the desired capacity defaults to your minimum capacity.

Elastic Load Balancing

the AWS service that automatically distributes incoming application traffic across multiple resources, such as Amazon EC2 instances.

A load balancer acts as a single point of contact for all incoming web traffic to your Auto Scaling group.

Although Elastic Load Balancing and Amazon EC2 Auto Scaling are separate services.



Monolithic application and microservices

Amazon Simple Notification Services (SNS)

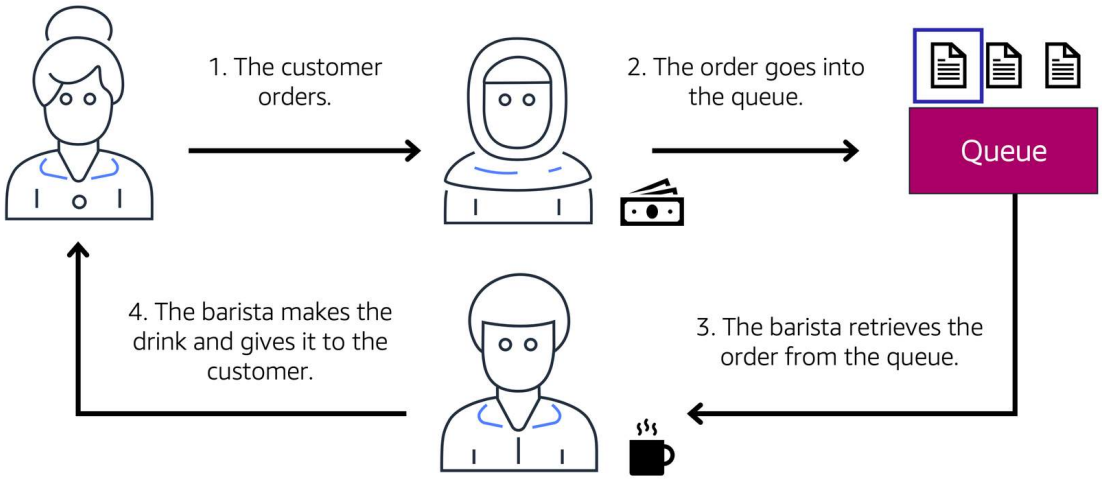
Using Amazon SNS topics, a publisher publishes messages to subscribers. This is similar to the coffee shop; the cashier provides coffee orders to the barista who makes the drinks.

In Amazon SNS, subscribers can be web servers, email addresses, AWS Lambda functions, or several other options.

Amazon Simple Queue Services(SQS)

Using Amazon SQS, you can send, store, and receive messages between software components, without losing messages or requiring other services to be available.

Amazon SQS, an application sends messages into a queue. A user or service retrieves a message from the queue, processes it, and then deletes it from the queue.

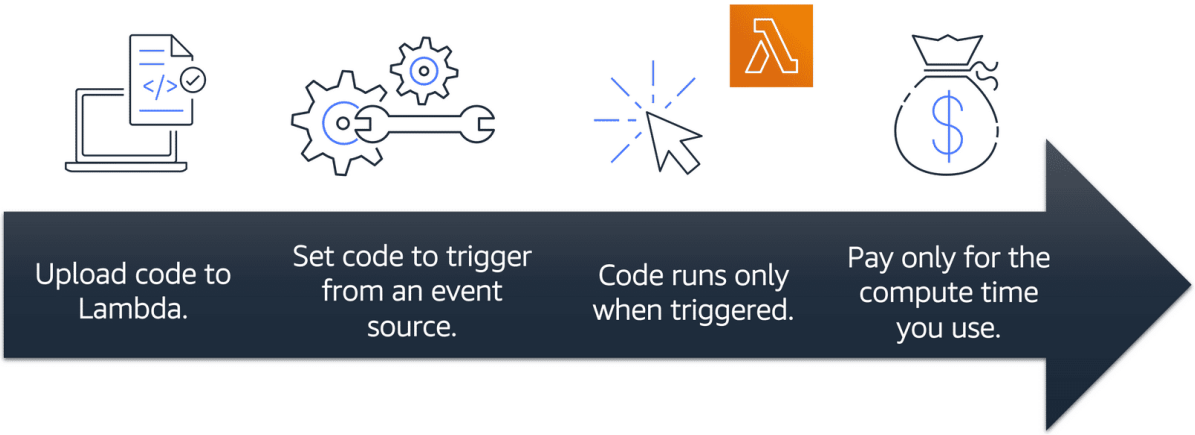


Serverless computing

The term “serverless” means that your code runs on servers, but you do not need to provision or manage these servers.

AWS Lambda

service that lets you run code without needing to provision or manage servers.



- set your code to trigger from an event source, such as AWS services, mobile applications, or HTTP endpoints.

Containers

Containers provide you with a standard way to package your application's code and dependencies into a single object.

Amazon Elastic Container Services (ECS)

highly scalable, high-performance container management system that enables you to run and scale containerized applications on AWS.

Amazon ECS supports Docker containers.

Amazon Elastic Kubernetes Services (EKS)

fully managed service that you can use to run Kubernetes on AWS.

Kubernetes is open-source software that enables you to deploy and manage containerized applications at scale.

You can also use containers for processes and workflows in which there are essential requirements for security, reliability, and scalability.

AWS Fargate

serverless compute engine for containers. It works with both Amazon ECS and Amazon EKS.

When using AWS Fargate, you do not need to provision or manage servers. AWS Fargate manages your server infrastructure for you.